

Genetics and Molecular Biology Midterm Exam

Question 1 (25 points)

Explain the difference between DNA replication and transcription. Include in your answer:

- The purpose of each process
 - The enzymes involved
 - The products generated
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Question 2 (25 points)

In pea plants, tall (T) is dominant to short (t) and yellow seeds (Y) are dominant to green seeds (y).

- a) If a heterozygous tall, heterozygous yellow plant is crossed with a short, green plant, what is the probability of getting a short plant with yellow seeds?
- b) What percentage of the offspring will be homozygous recessive for both traits?
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Question 3 (25 points)

Describe the process of protein synthesis from DNA to functional protein. Be sure to include:

- Transcription (including initiation, elongation, and termination)
 - RNA processing
 - Translation (including the roles of mRNA, tRNA, and ribosomes)
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Question 4 (25 points)

A DNA sequence is: 5'-ATGCGCTA-3'

- a) What is the complementary DNA strand?
- b) What is the mRNA transcript that would be synthesized from the template strand?
- c) What is the amino acid sequence that would be translated from this mRNA? (Use the genetic code table provided on the last page)
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Bonus Question (10 points)

Explain how mutations can lead to genetic variation and discuss one mechanism by which cells repair DNA damage.

Genetic Code Table Reference:

UUU, UUC = Phe | UUA, UUG = Leu | CUU, CUC, CUA, CUG = Leu

AUU, AUC, AUA = Ile | AUG = Met (start) | GUU, GUC, GUA, GUG = Val

UCU, UCC, UCA, UCG = Ser | CCU, CCC, CCA, CCG = Pro

ACU, ACC, ACA, ACG = Thr | GCU, GCC, GCA, GCG = Ala